

Combine Errors and Population Regression Line

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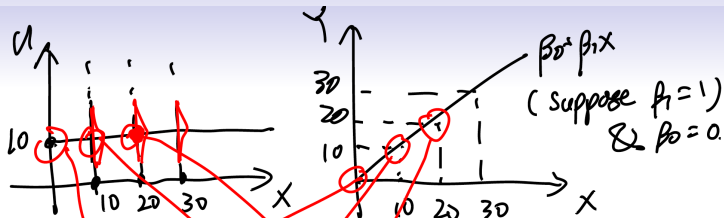
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- The derivation below is only for reference. You just need to remember: **we can always assume that $E[u] = 0$ holds** (but not necessarily for $E[u|X]$). **No need to recite the derivation.**
- We can show that $E[u] = 0$ always holds by re-defining the model. Suppose that for the model $Y = \beta_0 + \beta_1 X + u$, $E[u] = 10$ initially. We can re-define the model as follows:

$$\begin{aligned} Y &= (\beta_0 + 10) + \beta_1 X + (u - 10) \\ &= \beta_0^{new} + \beta_1 X + u^{new}, \end{aligned} \tag{1}$$

by letting $\beta_0^{new} = \beta_0 + 10$ and $u^{new} = u - 10$.

- Then, $E[u^{new}] = 0$. That is to say, **we can always make the unconditional mean of the error terms equals zero** (which means we can also assume that the unconditional mean of the errors equals zero in our linear regression model).
- Graphically, we have shifted the error u down by 10 while shifted the population regression line $\beta_0 + \beta_1 X$ up by 10. **The resulting new model will be totally equivalent to the original model.**



Combine

For $X=0$,
 combine 10 from U
 and 0 from Y
 $\Rightarrow 10$

For $X=10$
 10 from U
 10 from Y
 $\Rightarrow 20$

For $X=20$
 10 from U
 20 from Y
 $\Rightarrow 30$.